Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY **9744/03**
HIGHER 2 **23 August 2018**
Paper 3 Long Structured and Free-response Questions **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Writing Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show working or if you do not use appropriate units.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4 or 5	
TOTAL	**75**

Section A

Answer **all** the questions in this section.

1 Variant Creutzfeldt-Jakob disease (vCJD) is a rare infectious disease in humans that was first described in 1996 in the United Kingdom. It is thought to be caused by the same infectious agent which causes "mad cow" disease in cattle populations, where infected cows experience progressive deterioration of the nervous system and eventual death.

The infectious agent is a misfolded form of the protein PrP, which is normally found in nerve cells. This misfolded PrP protein is known as a prion, which can be transmitted to humans when eating food contaminated with these prions. In the brain, prions may bind to normal PrP proteins and induce them to change their conformation, forming more infectious prions which propagate the disease. Prions will bind to each other to form protein aggregates which are resistant to degradation, causing the death of nerve cells.

(a) Describe how an infectious disease differs from a genetic disease.

.... [2]

(b) Discuss how prions challenge the tenets of the cell theory.

.... [3]

(c) Suggest how protein aggregates may be resistant to degradation in infected cells.

.... [2]

While vCJD is described as an infectious disease, scientists have noted that there is little or no immune response in patients diagnosed with vCJD.

(d) Suggest why there is a lack of a significant immune response in these patients.

[2]

Similar to the prions causing vCJD, the human immunodeficiency virus (HIV) is thought to have originated from a related strain of viruses which infects wild chimpanzees.

A HIV infection is characterised by a brief initial period of viral replication. This is followed by a long period of latency, where no symptoms are manifested. Eventually, reactivation of the provirus and rapid viral replication results in immunodeficiency.

Progression of the disease in HIV-positive individuals can be monitored by measuring the concentration of HIV particles (also known as the viral load) in their blood sample. One laboratory method measures the viral load by quantifying the activity of viral reverse transcriptase using the following procedure:

1. The HIV particles are separated from other blood plasma components.
2. The viral particles are lysed by disruption of the viral envelope and the resulting lysates are added to a mixture containing a RNA template and deoxyribonucleotides.
3. Viral reverse transcriptase found in the lysate will then synthesise complementary DNA (cDNA).
4. The amount of cDNA formed is detected by adding artificially synthesised antibodies which recognise and bind to the cDNA. These antibodies are linked covalently to an enzyme, which catalyses the formation of a coloured product.
5. The colour intensity of the resulting mixture is compared to those of standard solutions, from which the concentration of reverse transcriptase in the starting mixture can be deduced.

(e) Explain how the structure of antibodies allow them to be used for the procedure described.

[3]

Antibodies are produced naturally by B lymphocytes in the human body, after exposure to foreign antigens.

(f) Justify the claim that "B lymphocytes are exceptions to the generalisation that all nucleated cells in the body have exactly the same DNA".

[4]

Another method to measure the HIV viral load involves quantifying the amount of viral genetic material present in the blood sample, using the procedure in Fig. 1.1.

HIV RNA extracted from blood plasma

↓

Artificial reverse transcriptase added, converting viral RNA into cDNA

↓

cDNA amplified using polymerase chain reaction (PCR) over a fixed number of cycles, using a primer which targets part of the viral sequences

↓

Amplified products used in further analysis to quantify genetic material present in initial blood sample

Fig. 1.1

(g) Describe how PCR is used in the amplification of HIV cDNA.

[4]

There are two major strains of HIV known to man, HIV-1 and HIV-2. The different methods used to measure viral load vary in their ability to distinguish between the different HIV strains, as shown in Table 1.1.

Table 1.1

Method of measuring viral load	**Able to distinguish between HIV strains**
Quantification of viral reverse transcriptase activity	No
Quantification of viral genetic material	Yes

(h) Explain the difference in the two measurement methods in their ability to detect different HIV strains.

[3]

While HIV is sexually-transmitted, dengue virus is transmitted by a mosquito vector. To study the influence of weather conditions on the incidence of dengue viral infection in Singapore, scientists recorded the amount of rainfall, mean weekly temperatures and the number of dengue cases over a one-year period. The recorded data is shown in Fig. 1.2.

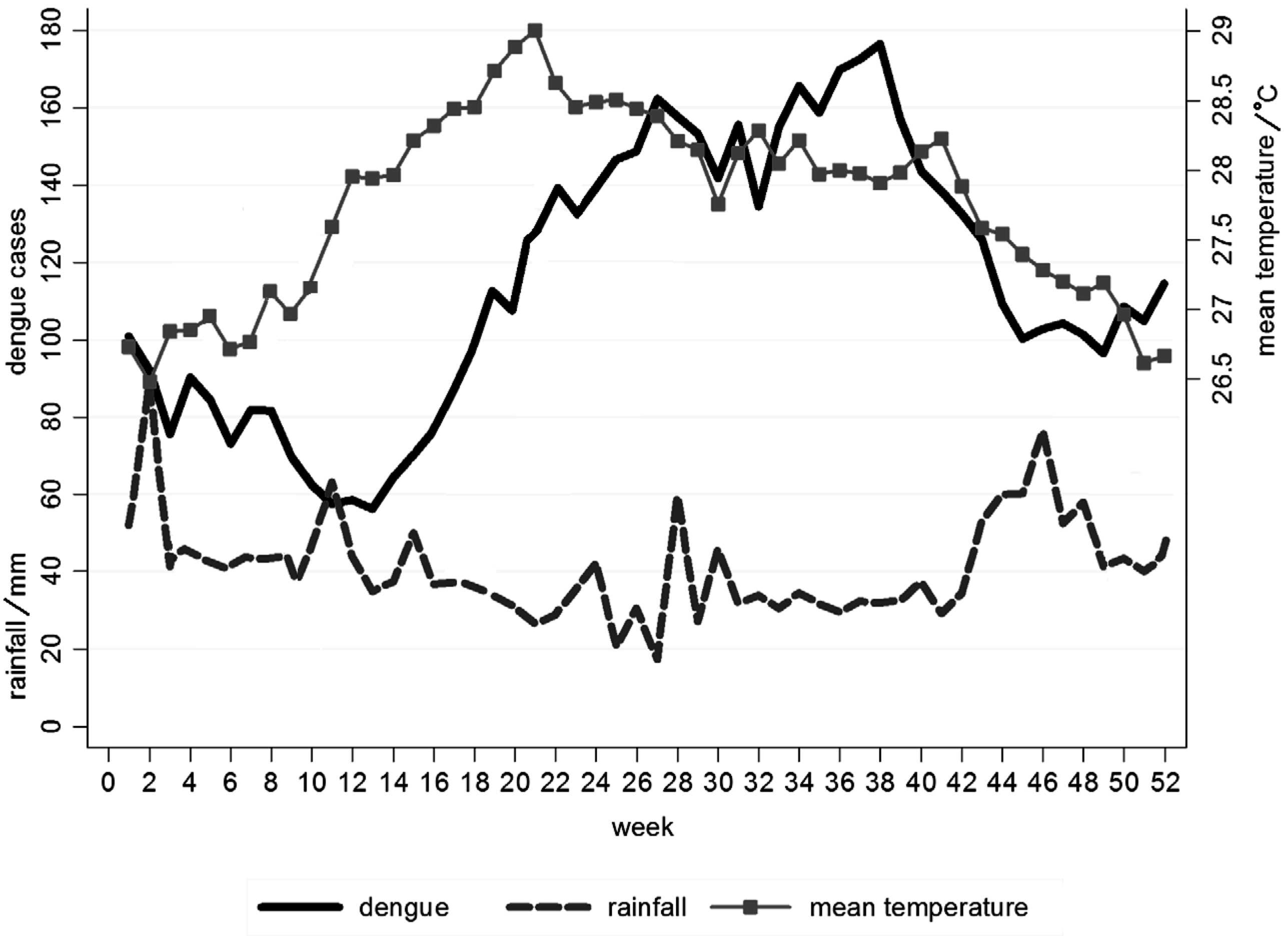

Fig. 1.2

(i) Based on the information provided, comment on which weather condition is a better predictor of dengue cases in Singapore.

[4]

(j) Suggest explanations for the relationship between the weather condition chosen in part (i) and the number of dengue cases recorded.

[2]

[Total: 29]

2 In order to solve the problems of malnutrition and nutritional deficiencies that are faced by two billion people worldwide, scientists have turned to genetic engineering. This involves the introduction of foreign genes into organisms to increase food production. One technique used in plant biotechnology to generate transgenic crops (crops with foreign genes introduced) involves the use of protoplast culture. It is designed to work as follows:

1. Cut a leaf into pieces.
2. Soak the leaf in a solution of microbial enzymes, sugars and salts. The enzymes digest the cellulose cell wall. The sugars and salts maintain the osmotic potential.
3. Centrifuge the cells to remove the cell wall debris. The protoplasts (plant cells with cell wall removed) will be found in the supernatant.
4. Apply an electric field to introduce the foreign genes into the protoplasts via transformation.
5. Isolate transformed protoplasts and culture them into transgenic plants.

(a) (i) Before a foreign gene is introduced into the protoplasts, the photosynthetic activity of the protoplasts must be checked to ensure that they are viable.

One way to do this is to mix the protoplast extract with 2,6-dichlorophenolindophenol (DCPIP), a redox dye. When reduced, DCPIP is changed from blue to colourless.

A protoplast mixture with DCPIP is placed under unfiltered white light, while another mixture is placed under white light passed through a blue filter.

Assuming that light intensity of the white light is constant, explain whether the DCPIP in the protoplast extract would decolourise faster under the unfiltered or filtered white light.

[4]

(ii) Suggest why it is necessary to digest the cellulose cell wall to allow for the introduction of a foreign gene into the protoplast.

[1]

Using the protoplast culture technique, scientists have introduced three β-carotene biosynthesis genes into a particular variety of the rice plant to form a new variety – the golden rice plant. The golden rice plant looks morphologically similar to other rice varieties except that its rice grains are yellow in colour due to the production of β-carotene in its rice grains. The β-carotene contained in the rice grains will be converted into vitamin A when it is consumed by humans.

(b) Discuss whether the β-carotene biosynthesis genes will be preserved in the gene pool of a population of rice plants, containing both the original and the new varieties, without any human intervention.

[4]

Genetic engineering can also be used to treat genetic diseases in humans which are caused by mutations involving single genes. This is done by introducing copies of the normal allele into human cells. In germline cell therapy, the normal alleles are introduced into germ cells.

(c) Comment on the ethical aspects of germline cell therapy.

[2]

[Total: 11]

3 Plants have long been regarded as carbon sinks because they take in carbon dioxide for photosynthesis. However, when temperatures rise, plants increase their rate of respiration, resulting in increased carbon release. Some research has suggested that this could convert forests from a long-term carbon sink to a carbon source, aggravating climate change.

(a) Outline how a rise in temperature could lead to increased carbon release by plants.

[3]

In 2016, a team of scientists conducted a short-term study of five years to find out the net carbon exchange of trees when the temperature was increased. In order to determine this, the increase in leaf respiration at higher temperatures was evaluated using 1000 young trees of 20 different boreal and temperate tree species grown in an open-setting.

Scientists compared the data observed to the expected results in Fig. 3.1.

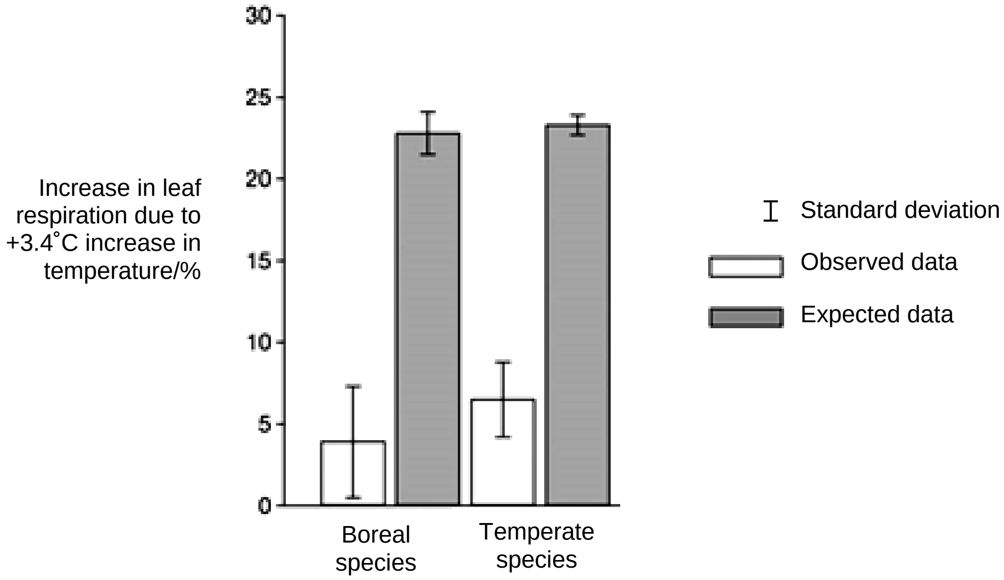

Fig. 3.1

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(b) **(i)** With reference to Fig. 3.1, describe one difference between the data observed and the expected results.

[2]

(ii) In Fig. 3.1, the data observed shows a difference in the increase in leaf respiration between boreal and temperate tree species. Explain whether this difference is significant.

[1]

(iii) Based on the results of the study, comment on whether forests will remain as carbon sinks or be converted to carbon sources if temperatures rise.

[2]

(iv) Based on the results, scientists concluded that plants can acclimatise to changing climates, adjusting their metabolism according the environment. It was thus suggested that crop yield and hence food supply would not change drastically in light of climate change.

With reference to the design of the experiment, discuss the validity of this conclusion.

[2]

[Total: 10]

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Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 Biological polymers are large molecules composed of many individual smaller molecules called monomers. For instance, amino acids are taken into an animal cell for the synthesis of collagen. The synthesised collagen is then secreted out of the cell to form the connective tissue.

(a) Explain how the structural features of amino acids contribute to the function of collagen. [10]

(b) Explain how temperature and pH may affect the transport of substances across the cell surface membrane of eukaryotes. [15]

[Total: 25]

5 There are similarities in the ways that prokaryotes and eukaryotes regulate gene expression. In both groups of organisms, cyclic adenosine monophosphate (cAMP) is an important molecule in many biological processes, including the regulation of gene expression.

(a) Describe the role of cAMP in altering gene expression in organisms and suggest why they are usually small and non-protein in nature. [10]

(b) The eukaryotic genome is considered to be more complex than the prokaryotic genome. Describe the differences between the eukaryotic and prokaryotic genomes and explain the significance of these differences. [15]

[Total: 25]